



ALPHA ARTS AND SCIENCE COLLEGE PORUR, CHENNAI -600 116

A PROJECT REPORT ON

LEGAL EASE AI

Powered Legal Document Generator

Submitted in partial fulfillment of the requirements under the
NAAN MUDALVAN (NM) SKILL DEVELOPMENT PROGRAMME

SUBMITTED BY

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1. INTRODUCTION

LegalEase is an AI-assisted legal document drafting application. It helps users create basic legal document drafts by entering important information such as document type, parties involved, terms and conditions, and effective date.

The system uses Streamlit for the frontend, FastAPI for the backend, and Gemini AI for generating the document draft. The generated document can be reviewed, edited, and downloaded in different formats.

2. PROBLEM STATEMENT

Preparing a structured legal document manually can take time and requires repeated formatting. Users may find it difficult to organize parties, dates, terms, responsibilities, and signature sections properly.

LegalEase solves this problem by collecting the required information from the user and generating a structured legal document draft that can be reviewed and edited.

3. OBJECTIVES

The main objectives of LegalEase are:

- To create an easy-to-use interface for entering legal document details.
- To generate structured legal document drafts using Gemini AI.
- To connect the Streamlit frontend with a FastAPI backend.
- To allow users to review and edit generated content.

- To provide document export in TXT, DOCX, and PDF formats.
- To avoid inventing missing names, dates, amounts, or other facts

4. EXISTING SYSTEM

In the existing manual system, users prepare legal documents themselves using word-processing software or templates.

This process may involve:

- Repeated typing
- Formatting headings
- Organizing clauses
- Adding party details
- Creating different document formats

The process can therefore require more time and manual effort.

5. PROPOSED SYSTEM

LegalEase provides a single workflow for creating legal document drafts.

The user enters the required information through the Streamlit interface. The information is sent to the FastAPI backend, which communicates with Gemini AI. Gemini generates a structured document draft and sends it back to the frontend.

The user can then review, edit, and download the document.

6. SYSTEM REQUIREMENTS

Hardware Requirements

- Computer or Laptop

- Internet Connection
- Sufficient storage space
- Keyboard and Mouse

Software Requirements

- Python
- Visual Studio Code
- Streamlit
- FastAPI
- Uvicorn
- Google Gemini API
- Microsoft Word or compatible document viewer

7. TECHNOLOGIES USED

Technology	Purpose
Python	Main programming language
Streamlit	Frontend user interface
FastAPI	Backend API
Uvicorn	Backend server
Gemini AI	AI document generation
Pydantic	Input validation
python-docx	DOCX generation
ReportLab	PDF generation
Pillow	Image and logo handling
Requests	Frontend-backend communication

Technology

Purpose

python-dotenv Configuration management

8. SYSTEM ARCHITECTURE

The main architecture of LegalEase is:

User → Streamlit Frontend → FastAPI Backend → Gemini AI → Generated Draft → Edit → Export

The frontend collects the required information. FastAPI validates and processes the request. Gemini AI generates the document draft. The generated result is returned to Streamlit for review and editing.

Finally, the document can be exported as TXT, DOCX, or PDF.

9. MODULES

9.1 User Input Module

This module collects:

- Document Type
- Parties Involved
- Terms and Conditions
- Effective Date

9.2 Backend API Module

This module receives the user's information and handles document generation and export requests.

9.3 AI Generation Module

This module uses Gemini AI to generate the legal document draft.

9.4 Editing Module

This module allows the user to review and edit the generated document.

9.5 Export Module

This module creates:

- TXT
- DOCX
- PDF

9.6 Validation Module

This module checks the required fields and input length before processing.

10. SYSTEM WORKFLOW

The workflow of LegalEase is:

1. Open the LegalEase application.
2. Enter the document type.
3. Enter the parties involved.
4. Enter the terms and conditions.
5. Enter the effective date.
6. Click **Generate Document**.
7. Streamlit sends the request to FastAPI.
8. FastAPI sends the information to Gemini AI.
9. Gemini generates the document draft.
10. The generated document is displayed.
11. The user reviews and edits the document.
12. The final document is downloaded as TXT, DOCX, or PDF.

11. IMPLEMENTATION

The backend of LegalEase is developed using FastAPI. Configuration values such as the Gemini API key and model settings are stored using environment variables.

Pydantic schemas are used to validate the incoming information. FastAPI routes handle document generation and export requests.

The frontend is developed using Streamlit. It provides fields for entering the document type, parties, terms, and effective date. After generation, the document is displayed in an editable text area.



AI-Assisted Legal Document Drafting

Create a legal document draft using your document details and terms.

Document Type

Example: Non-Disclosure Agreement

Parties Involved

Example: Party A: ABC Company
Party B: XYZ Company

Terms and Conditions

12. DOCUMENT GENERATION

The user provides the required document information. LegalEase sends these details to Gemini AI.

The AI is instructed to use only the information supplied by the user and not to invent names, dates, addresses, money amounts, laws, courts, or other facts.

If important information is missing, a placeholder such as **[INSERT INFORMATION]** can be used.

The generated document contains suitable headings and numbered clauses where applicable.

Document Type

Non-Disclosure Agreement

Parties Involved

Party A

Terms and Conditions

Enter the terms and conditions...

Effective Date

Example: 24 September 2026

Generate Document

13. EDITING THE DOCUMENT

After the document is generated, the user can review the content and make necessary changes.

The editable document area allows the user to modify the generated text before downloading the final document.

Generated Document

Edit your document

3.1 "Confidential Information" shall mean [INSERT DETAILED DEFINITION OF CONFIDENTIAL INFORMATION OR SCOPE OF INFORMATION COVERED].

4. MAIN TERMS AND CONDITIONS

4.1 Mutual Protection: Both parties agree to keep confidential information private.

4.2 Standard of Care: Each Party agrees to protect the Confidential Information of the other Party with at least the same degree of care it uses to protect its own confidential information, but in no event less than a reasonable degree of care.

5. RESPONSIBILITIES

5.1 Non-Disclosure: Neither Party shall disclose, publish, or otherwise make available the Confidential Information of the other Party to any third party without the prior written consent of the disclosing Party, except as permitted under [INSERT PERMITTED DISCLOSURES OR EXCEPTIONS, IF ANY].

5.2 Limited Use: Each Party shall use the Confidential Information solely for the purpose of [INSERT PURPOSE OF DISCLOSURE].

6. TERMINATION

6.1 Term: This Agreement and the confidentiality obligations herein shall remain in effect until [INSERT TERMINATION DATE OR EVENT] or for a period of [INSERT DURATION, E.G., NUMBER OF YEARS] from

14. EXPORT AND DOWNLOAD

LegalEase supports three output formats:

TXT

The document can be downloaded as a simple text file.

DOCX

The document can be downloaded as an editable Microsoft Word file.

PDF

The document can be downloaded as a formatted PDF file.

The backend also supports optional logo handling for DOCX and PDF generation.

15. CONCLUSION

LegalEase demonstrates an end-to-end AI-assisted document drafting workflow. It combines Streamlit, FastAPI, and Gemini AI to collect user information, generate a structured legal document draft, allow review and editing, and support document export.

The project provides a practical example of integrating a modern AI service with a Python web application.

GITHUB :

<https://github.com/kowsalyasmsk-web/LegalEase>

DEMO LINK :

https://drive.google.com/file/d/1Y_96oK77Z3eD90A5kfRIsHo0w8tkozDN/view?usp=drivesdk